Jules Couprie

Aerodynamics & CFD Engineering Intern

12-month industrial placement| Available from July 2026

Contact

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Portfolio - More detailed resume

Hard skills

- CAD - 3DX / Catia V5 / Solidworks
- CFD - Ansys suite / Paraview
- MATLAB / Simulink
- Python (Pycharm)
- Video editing - DaVinci Resolve

Soft skills

Project Management:
Structured engineering workflows, defined task milestones, managed project deadlines

Team Leadership
Coordinated cross-functional student teams to solve complex problems.

Technical Communication:
Delivered clear technical documentation, executive reports, and media content to diverse audiences.

Hobbies

Sports & Endurance
- Finisher of the GR20
- Trail running - marathon runner
- Climbing
- 8 years skateboarding

Media Production
- Videography & Photography
- Video editing (Davinci resolve)

Language

French - Native

English- Full professional proficiency
TOEIC 960 / Cambridge assessment

German- Limited working proficiency

Profile

Engineering student at ESTACA specialising in mechanics and advanced aeronautical technologies, with hands-on experience in CFD and aerodynamic design. Currently developing an F1 front wing to evaluate aerodynamic performance. Seeking an industrial placement in aerodynamic design / CFD engineering starting in July 2027. Highly motivated and adaptable, I bring a rigorous scientific approach and strong team spirit to every engineering challenge.

Experience

F1 Front Wing Aerodynamic & CFD (2026 – Present)

Supervision & Mentorship: *Supervised by ONERA • Supported by Louis Guidet (Aerodynamicist at Alpine F1 Team)*

- Defined a progressive design methodology, starting from a single-element airfoil and building toward a full FIA-compliant multi-element front wing with wheel-wake interaction.
- Modeling a fully parametric wing geometry in 3DEXPERIENCE (Generative Shape Design), to be validated against CFD in Ansys Fluent.
- Study plan targets total pressure coefficient, sidewash, and L/D ratio as key performance metrics.

ČVUT - Prague (July - August 2026)

Aero Research Intern

Simulation and initial design of peristaltic pump :

- **FSI Co-Simulation:** Configured two-way Fluid-Structure Interaction (FSI) to capture structural flexibility and fluid coupling.
- **Dynamic Mesh Setup:** Implemented dynamic remeshing in ANSYS to model moving geometry and large domain deformations.
- **Flow Analysis:** Developed Python post-processing for flow-rate and stagnation analysis

HYCE - iDrogen (2025 - Present)

Embedded Systems Member & Communications Manager

- Dimensioned and integrated a high-pressure hydrogen storage system for a hydrogen-retrofit vehicle targeting the Nürburgring lap record in our category.
- Produced technical media content (website, videos) to document and promote project progress.
- Also served as Secretary of the iDrogen ESTACA Club, coordinating meetings and monitoring action plans.

SD Aviation - La Roche-sur-Yon (July 2024)

Flight operator

- Managed flight planning and scheduling using Leon and ForeFlight software.
- Operational support: Managed an online marketplace and performed various technical and manual maintenance tasks. Compliance matrices and quality control

Education

ESTACA, Laval (2023- Present)

Master's Degree in Aeronautical Engineering

- Specialisation in mechanics and advanced aeronautical technologies
- UCAV Design project on CATIA (Strategic and technical analysis, aerodynamic design and dimensioning, CAD modeling and simulation)
- Courses: Mathematics, Fluid mechanics, Aerodynamics, System mechanics, SOLIDWORKS, CATIA/3DX, MATLAB